

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Machine Learning & Evaluation Aug 2025 to May 2026

- Boosted malware-family classification to **80.2%** accuracy across **17 families** by soft-voting **ResNet50**, **ViT-B/16**, VGG16, HOG, and Haralick models, beating the best single model by **2.4 pts** (McNemar $p < 0.001$).
- Proved a **2.4 pt** ensemble gain was real with **McNemar** testing at $p < 0.001$ instead of reporting a single accuracy number, across **1,700** held-out samples.
- Generated **136,000** training images from **17,000** binaries across **8** visual representations and benchmarked **6** model families on **1,700** held-out samples.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.

Turito | Software Engineering Intern, Systems & Reliability Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, TypeScript, SQL

ML & Retrieval: PyTorch, transformers, embeddings, RAG with citations and abstention, two-tower retrieval, HNSW and ANN search, semantic caching

Evaluation: LLM eval harnesses, McNemar significance testing, leakage-free temporal splits, Recall/NDCG/MAP, failure classification, human-review gates

Systems: FastAPI, Kafka, Temporal, gRPC, Docker, Kubernetes, AWS, Linux, CI/CD

PROJECTS

Teller, Agent Evaluation Environment | *Browser agents, Claude Code, Python, TypeScript*

- Analyzed **14** computer-use agent runs against an adversarial app (nested frames, missing test IDs, session expiry), classifying each failure as planning, execution, validation, or recovery.

AutoGrade AI, Multi-Agent Evaluation Platform | *Python, Temporal, Kafka, gRPC, PostgreSQL, Qdrant*

- Orchestrated **4-dimension** rubric grading (correctness, efficiency, resilience, quality) across multiple LLM agents on **Temporal** and **Kafka**, with confidence gates and human-review queues.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

OPEN SOURCE

- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)
- **microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Advanced Algorithms, Distributed Computing, Database Systems, GPU Programming

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)